

Understanding Metadata

A Beginner's Guide for Data Science

What is Metadata?

Metadata is simply "data about data." It provides the essential context—the who, what, when, where, and why—that transforms meaningless raw numbers or text into usable information.

Analogy: The label on a soup can (metadata) tells you what's inside the can (the data).

1. Types of Metadata

- **Descriptive:** Answers "What is this?" (e.g., dataset title, author, keywords). Helps in finding and identifying data.
- **Structural:** Answers "How is this built?" (e.g., data types like integers/strings, file formats).
- **Administrative:** Answers "How is this managed?" (e.g., creation dates, access permissions, data lineage).

2. The 5 Characteristics of Good Metadata

1. **Accuracy:** It must truthfully and precisely reflect the actual data (e.g., if metadata says "Celsius", the data cannot be in Fahrenheit).
2. **Completeness:** It leaves no gaps. All columns, null values, and data sources are fully explained.
3. **Consistency:** It uses standardized formatting and naming conventions across the entire organization.

4. **Timeliness:** It is updated immediately whenever the underlying data structure or pipeline changes.
5. **Accessibility:** It is easy for data scientists to search, find, and read (usually hosted in a centralized Data Catalog).

3. Implementation: Viewing Metadata in Pandas

Below is an example of how a data scientist interacts with metadata using Python and the Pandas library.

PYTHON CODE

```
import pandas as pd

# 1. THE RAW DATA
data = {
    'customer_id': [101, 102, 103],
    'age': [22, 28, 25],
    'signup_date': pd.to_datetime(['2025-01-15', '2025-02-20', '2025-03-10'])
}
df = pd.DataFrame(data)

# 2. ADDING DESCRIPTIVE & ADMINISTRATIVE METADATA
df.attrs['dataset_name'] = "Customer_Signup_Q1"
df.attrs['author'] = "Data Science Team"
df.attrs['last_updated'] = "2026-08-11"
df.attrs['description'] = "Ages and signup dates for early 2025 cohort."

# 3. EXTRACTING AND VIEWING THE METADATA
print("--- 1. STRUCTURAL METADATA ---")
print(df.dtypes)
print("\nDataFrame Shape:", df.shape)

print("\n--- 2. DESCRIPTIVE METADATA ---")
for key, value in df.attrs.items():
    print(f"{key.upper()}: {value}")
```